

Why the Heuristic Outperforms Meta-Heuristics on Large Graphs

The superior performance of the proposed heuristic on large-scale instances stems from two critical design decisions embedded in its construction logic. First, the algorithm integrates a physics-based, load-dependent energy model directly into its node selection criterion, evaluating each candidate based on a composite cost of normalized travel time and energy consumption. This stands in contrast to the Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Clarke-Wright (CW) approaches, which construct routes using time- or distance-based heuristics and defer energy considerations to a post-construction evaluation phase. Second, the heuristic employs a Breadth-First Search (BFS) reachability filter that prunes the candidate set to only those nodes reachable with the current battery state, eliminating infeasible options before any computational effort is spent evaluating them. Together, these mechanisms ensure that every routing decision is both energy-aware and guaranteed feasible, a property that the meta-heuristics cannot replicate without incurring prohibitive computational overhead.

On smaller problem instances, meta-heuristics such as GA and ACO can effectively explore the solution space through stochastic search, occasionally discovering globally optimal routes that a greedy heuristic might miss. However, as the number of customers grows, the solution space expands factorially, and the probability of random exploration yielding high-quality solutions diminishes sharply. The GA's crossover operator, which recombines partial routes, increasingly destroys good sub-structures as routes lengthen. Similarly, ACO's pheromone trails become diluted across a larger edge set, weakening the reinforcement signal for promising paths. Both algorithms also suffer from an explosion of infeasible solutions—routes that violate battery constraints—which consume evaluation resources without contributing to convergence. The proposed heuristic avoids these pitfalls entirely: its greedy, energy-aware decisions compound into consistently high-quality solutions, and its BFS filter

scales linearly with graph size rather than with the combinatorial explosion of possible routes.

Clarke-Wright exhibits scalability characteristics closest to the proposed heuristic because both are constructive, single-pass algorithms that avoid the population management overhead inherent in GA and ACO. The performance gap between them arises not from computational complexity but from the objective embedded in the construction step: CW optimizes for distance savings when merging routes, whereas the proposed heuristic optimizes for energy efficiency when selecting the next node. Since the Electric Vehicle Routing Problem is fundamentally constrained by battery capacity, the energy-aware objective produces routes that are better aligned with the problem's core challenge. General-purpose meta-heuristics, by contrast, treat energy as an external fitness metric rather than an integral construction guide, limiting their ability to adapt to the load-dependent dynamics of EV routing as problem scale increases.

This is an excellent question about how to frame your contribution. The answer is **both**:

It Is Their Inherent Limitation

Standard meta-heuristics face a fundamental barrier to embedding energy into construction:

Algorithm	Why Energy-Aware Construction Is Hard
GA	Crossover recombines route segments blindly. There is no "moment" to calculate energy because the full route is assembled at once.
ACO	Pheromones are deposited on <i>edges</i> . But energy cost of edge (A→B) depends on <i>current load</i> , which varies. You can't assign a single pheromone value to an edge when its "goodness" changes depending on when it's traversed.
CW	Savings are computed <i>before</i> construction starts. At that point, you don't know what order customers will be visited, so you can't know the load at each merge point.

This is a **structural limitation** of these paradigms for state-dependent problems like load-dependent EVRP.

It Is Your Achievement

Your heuristic **overcomes** this limitation because of its constructive, step-by-step nature:

- At each step, you **know** the exact vehicle state (battery level, current load)
- You can **calculate** the precise energy cost to each candidate with that state
- You can **filter** unreachable candidates before wasting computation

This is not trivial — you designed an algorithm that exploits the structure of the problem in a way that general-purpose meta-heuristics cannot.

How to Frame This in Your Paper

"A fundamental challenge in applying population-based meta-heuristics to load-dependent EVRP is that energy consumption is a state-dependent quantity: the energy cost of traversing an edge varies with the vehicle's current payload. Standard implementations address this by evaluating energy during route simulation, after the route structure is determined, rather than during construction. This decoupling limits the algorithm's ability to make energy-informed routing decisions. The proposed heuristic overcomes this limitation by leveraging its constructive nature: at each decision step, the algorithm knows the precise vehicle state and can evaluate candidate nodes using an accurate, physics-based energy model. This tight integration of energy into the construction process—rather than relegating it to post-hoc evaluation—constitutes a key contribution of this work."

In short: Their limitation is real (structural). Your achievement is finding a design that sidesteps it.

Correct. Even if you made the objective function identical, they would still NOT perform the same. Here's why:

The Objective Function vs. The Construction Heuristic

	Objective Function (Fitness)	Construction Heuristic
Purpose	Scores a <i>complete</i> route	Guides <i>how</i> the route is built
When used	After route is constructed	During route construction

If you change GA/ACO/CW to use

$$\text{cost} = 0.5 * \text{time} + 0.5 * \text{energy}$$

:

Algorithm	What Changes	What Stays the Same
GA	Fitness function uses energy	Crossover still recombines routes randomly (no energy guidance)
ACO	Final cost uses energy	Attractiveness heuristic is still $\eta = 1/\text{time}$ (no energy guidance)
CW	Route cost uses energy	Savings calculation is still distance-based (no energy guidance)

Analogy

Imagine you're building a house:

- **Objective function** = How you *judge* the house after it's built (price, quality)
- **Construction heuristic** = How you *choose* each brick while building

Changing the objective is like saying "I'll judge the house by energy efficiency" — but if the workers are still placing bricks randomly, the house won't be energy-efficient just because you *want* it to be.

Your heuristic's advantage is that **the workers (the algorithm) choose each brick (node) specifically for energy efficiency.**

What Would Happen If You Did This

Algorithm	With Energy Objective	Result
GA	$\text{Fitness} = 0.5\text{time} + 0.5\text{energy}$	Selection pressure would favor energy-efficient routes, BUT crossover still randomly destroys/creates routes. Might improve slightly, but construction is still blind.
ACO	$\text{Cost} = 0.5\text{time} + 0.5\text{energy}$	Pheromones would eventually reward energy-efficient routes, BUT the attractiveness heuristic ($1/\text{time}$) still guides ants toward fast edges, not energy-efficient ones.
CW	$\text{Cost} = 0.5\text{time} + 0.5\text{energy}$	Merged routes are scored better, BUT the savings calculation (which drives merging order) is still distance-based.

The Bottom Line

Changing the objective function makes the algorithm recognize good solutions but doesn't make it construct good solutions.

Your heuristic does both:

1. **Constructs** energy-aware routes (BFS filter + energy in selection)
2. **Recognizes** energy-aware routes (the same cost function)

This is why your approach is fundamentally different, not just a parameter change.